Question 1

Create three simple scripts and name them Linux4Hackers1, Linux4Hackers2, Linux4Hackers3

Approach

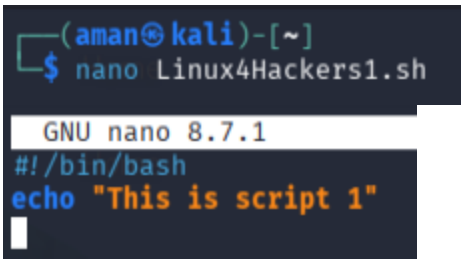
I created three small bash scripts similar to previous chapter script and saved them with the required filenames.

Commands

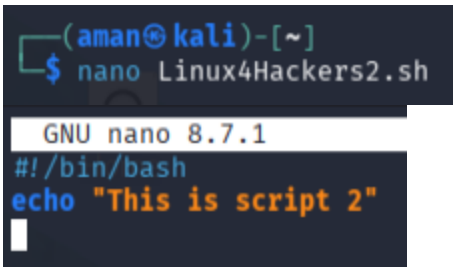
- `nano` → create script
- `echo` → display message
- `chmod +x` → make executable

Output

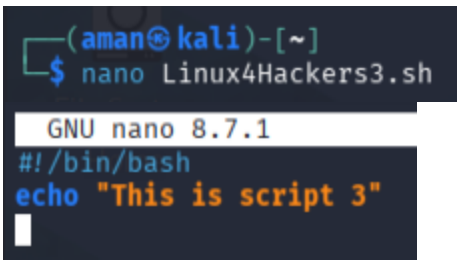
STEP 1 : create first script



STEP 2 : create second script



STEP 3 : create third script



STEP 4 : make scripts executable

```
┌──(aman㉿kali)-[~]
└─$ chmod +x Linux4Hackers1.sh Linux4Hackers2.sh Linux4Hackers3.sh
```

Key Observations

Multiple scripts can be created and prepared for combining

What I learned

Scripts can be grouped for packing or transfer.

Question 2

Create tar archive named L4H from the three scripts

Approach

I combined the three script files into one tar archive file.

Commands

- `tar` → combine files
- `-c` → create archive
- `-f` → specify filename

Output

STEP 1 : create tarball

```
┌──(aman㉿kali)-[~]
└─$ tar -cvf L4H.tar Linux4Hackers1.sh Linux4Hackers2.sh Linux4Hackers3.sh
Linux4Hackers1.sh
Linux4Hackers2.sh
Linux4Hackers3.sh
```

STEP 2 : verify file

```
┌──(aman㉿kali)-[~]
└─$ ls -lh L4H.tar
-rw-rw-r-- 1 aman aman 10K Mar 31 05:25 L4H.tar
```

Key Observations

Tar combines multiple files into single archive.

What I learned

Tar simplifies file transfer and storage.

Question 3

Compress L4H archive using gzip and then decompress

Approach

I compressed the tar file using gzip and then restored it.

Commands

- `gzip` → compress file
- `gunzip` → decompress file

Output

STEP 1 : compress archive

```
┌──(aman㉿kali)-[~]
└─$ gzip L4H.tar
```

STEP 2 : check compressed file

```
┌──(aman㉿kali)-[~]
└─$ ls
Desktop  Documents  Downloads  greeting.sh  L4H.tar.gz
```

STEP 3 : decompress file

```
┌──(aman㉿kali)-[~]
└─$ gunzip L4H.tar.gz
```

Key Observations

Compression reduces file size.

What I learned

gzip helps reduce storage and speeds transfer.

Question 4

Compress L4H archive using bzip2 and compress command

Approach

I used two different compression tools to compare results.

Commands

- `bzip2` → stronger compression
- `compress` → traditional compression

Output

STEP 1 : compress using bzip2

```
┌──(aman㉿kali)-[~]
└─$ bzip2 L4H.tar
┌──(aman㉿kali)-[~]
└─$ ls
Desktop  Documents  Downloads  greeting.sh  L4H.tar.bz2
```

STEP 2 : decompress bzip2 file

```
┌──(aman㉿kali)-[~]
└─$ bunzip2 L4H.tar.bz2
┌──(aman㉿kali)-[~]
└─$ ls
Desktop  Documents  Downloads  greeting.sh  L4H.tar
```

STEP 3 : compress using compress tool

```
┌──(aman㉿kali)-[~]
└─$ compress L4H.tar
┌──(aman㉿kali)-[~]
└─$ ls
Desktop  Documents  Downloads  greeting.sh  L4H.tar.Z
```

STEP 4 : decompress file

```
┌──(aman㉿kali)-[~]
└─$ uncompress L4H.tar.Z

┌──(aman㉿kali)-[~]
└─$ ls
Desktop  Documents  Downloads  greeting.sh  L4H.tar
```

Key Observations

Different tools produce different compression ratios.

What I learned

Multiple compression methods exist for different use cases.

Question 5

Create bit-by-bit copy of flash drive using dd command

Approach

I used dd command to create an exact binary copy of a storage device.

Commands

- `dd` → disk duplication tool
- `if` → input file/device
- `of` → output file

Output

STEP 1 : identify drive

```
┌──(aman㉿kali)-[~]
└─$ lsblk
NAME   MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda      8:0    0 30.7G  0 disk
├─sda1   8:1    0 29.1G  0 part /
├─sda2   8:2    0    1K  0 part
└─sda5   8:5    0  1.6G  0 part [SWAP]
sr0     11:0    1 1024M  0 rom
```

STEP 2 : create disk image

```
┌──(aman㉿kali)-[~]
└─$ sudo fdisk -l
Disk /dev/sda: 30.69 GiB, 32952942592 bytes, 64361216 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x58a184be

Device     Boot    Start      End  Sectors  Size Id Type
/dev/sda1  *        2048 60971007 60968960 29.1G 83 Linux
/dev/sda2       60973054 64360447  3387394  1.6G  f W95 Ext'd (LBA)
/dev/sda5       60973056 64360447  3387392  1.6G 82 Linux swap / Solaris
```

```
┌──(aman㉿kali)-[~]
└─$ sudo dd if=/dev/sda1 of=flashcopy.img bs=4M status=progress
14516486144 bytes (15 GB, 14 GiB) copied, 860 s, 16.9 MB/s
dd: error writing 'flashcopy.img': No space left on device
3462+0 records in
3461+0 records out
14516486144 bytes (15 GB, 14 GiB) copied, 864.176 s, 16.8 MB/s
```

Key Observations

dd copies entire disk including deleted data.

What I learned

dd creates forensic-level disk images.

Security Insights

1. Attackers often bundle tools into archives for stealth transfer.
2. Compressed files help hide malicious payloads.
3. gzip archives may contain hidden scripts or malware.
4. Strong compression helps evade detection systems.
5. dd can copy sensitive data including deleted files.